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On the Stability and Scalability of Vision Transformers with Manifold-Constrained Hyper-Connections

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THERE IS HIGH LIKELIHOOD THAT AI WAS USED IN WRITING THIS. PLEASE REFRAIN FROM THE USE OF AI FOR THE FINAL REPORT AND FOR CONDUCTING THE PROJECT.

I. Project Description

Vision Transformers (ViTs) have emerged as a dominant architecture in computer vision due to their scalability and strong empirical performance across classification, detection, and segmentation tasks [1]–[3]. By leveraging global self-attention mechanisms and deep residual structures, ViTs effectively model long-range dependencies and scale to large model sizes.

Despite their success, training very deep or large-scale ViTs remains challenging. As model depth and width increase, optimization can become unstable due to representation drift, gradient degradation, and sensitivity to residual mixing dynamics [4]. The stability of residual signal propagation plays a critical role in enabling scalable transformer training [5].

Recent work has proposed Hyper-Connections (HC) [6], which extend conventional residual blocks into multi-stream mixing structures to increase representational flexibility. By widening the residual pathway and allowing learnable cross-stream interactions, HC improves representational capacity beyond standard identity shortcuts. However, unconstrained mixing may disrupt identity-like propagation and introduce instability in deep networks.

Manifold-Constrained Hyper-Connections (mHC) [7] address this limitation by constraining residual mixing matrices to lie on a doubly stochastic manifold. This constraint preserves convex signal propagation and maintains identity-like behavior while retaining the expressive benefits of multi-stream residual structures. By projecting mixing matrices onto a structured manifold, mHC stabilizes gradient flow and improves conditioning in deep transformer architectures.

Subsequent extensions such as KromHC [8] and mHC-lite [9] further improve computational efficiency and structural properties, making manifold-constrained residual mixing more practical for large-scale vision models.

While mHC has demonstrated improved training stability in large language models and transformer-based architectures, its application to Vision Transformers has not yet been thoroughly investigated. In this work, we explore integrating mHC into ViT architectures to evaluate its impact on optimization stability, scalability, and downstream vision performance.

II. Review of the State of the Art

A. Vision Transformers and Training Stability

Vision Transformers (ViTs) have emerged as a dominant architecture in computer vision due to their scalability and strong empirical performance across classification, detection, and segmentation tasks [1]–[3]. By leveraging global self-attention mechanisms and deep residual structures, ViTs effectively model long-range dependencies while scaling to large model sizes.

Despite their success, training ViTs from scratch historically required large-scale pretraining and careful optimization. Previous work has demonstrated that with appropriate regularization, augmentation strategies, and optimization techniques, ViTs can be trained effectively without massive pretraining datasets [5]. Subsequent architectural modifications, such as Tokens-to-Token ViT [10], improved local structure modeling and enhanced data efficiency. Other efforts, including pyramid adversarial training [11], further improved robustness and performance. Previous work centered around promoting stability of transformers found that normalization was crucial for stabilization [12] but stabilization as a whole has not had recent extensive work done.

Recent comparative studies of CNN–ViT hybrids [13] highlight ongoing trade-offs between computational efficiency, inductive bias, and scalability. While these works focus on architectural inductive biases and training strategies, fewer studies examine the role of residual mixing dynamics in transformer stability.

B. Hyper-Connections and Manifold-Constrained Extensions

Hyper-Connections (HC) extend standard residual connections into multi-stream mixing architectures, increasing representational flexibility [6]. However, unconstrained residual mixing may disrupt identity-like signal propagation and lead to optimization instability in deep networks.

Manifold-Constrained Hyper-Connections (mHC) address this limitation by constraining residual mixing matrices to lie on a doubly stochastic manifold [7]. This constraint preserves convex signal propagation and stabilizes gradient flow while retaining expressive capacity. Recent work has applied mHC to graph neural networks [14], other work improves on the mathematical foundations of mHC by improving computational efficiency and structural properties such as KromHC [8], and mHC-lite [9].

Although mHC has demonstrated stability improvements in transformer-based architectures, its application to Vision Transformers remains largely unexplored.

III. Proposed Method and Research Roadmap

A. Proposed Method

We propose to integrate mHC into standard ViT architectures by replacing conventional residual connections in transformer encoder blocks with manifold-constrained multi-stream residual structures. Specifically, residual mixing matrices will be constrained to be doubly stochastic using iterative normalization procedures as described in [7].

The following model variants will be evaluated:

- ViT with no modifications
- ViT + established architectural extensions (Tokens-to-Token and or pyramid adversarial training)
- ViT + mHC (potentially also could evaluate HC)
- ViT + mHC-lite or KromHC (depending on resources, both extensions may not be necessary as they would likely yield similar results)

Depending on the availability of resources, the ViT models have different variants as well which could multiply the amount of models that can be evaluated. The mHC effect may not change drastically between ViT models so settling on one ViT model with proven accuracy / instability (consult previous research) would likely be the best. One or two extensions of ViT that increase accuracy / stability would be good to check mHC against previous work. For completion sake the two extensions of mHC-lite and KromHC should be evaluated but if resources are limited, one extension is enough.

B. Research Roadmap

Phase 1: Baseline Replication Reproduce base ViT training from scratch following established training protocols [5]. Reproduce an extension of ViT from either Tokens-to-Token [10] or Pyramid Adversarial training [11].

Phase 2: mHC Integration Implement mHC residual mixing within ViT blocks and evaluate training dynamics under identical hyperparameters [7]. Implement an extension of mHC from either KromHC [8] or mHC-lite [9].

Phase 3: Experimentation If we want to have multiple ViTs, we could evaluate shallow and deeper configurations (e.g., 12-layer vs. 24-layer ViTs) to assess stability and depth scalability. Run experiments on several datasets, keeping track of computational overhead, memory usage, time to train, accuracy, F1, precision, recall, etc. For stability purposes, we can keep track of metrics like training loss, gradient norms, attention scores, and weight norms to monitor how the ViT learns.

Phase 4: Result Interpretation & Writing Compiling all the results into useful visualizations. Cleanup writing for background, methodology, results, and discussion into hopefully a manuscript that can be published.

IV. Proposed Data

The project will use publicly available image classification datasets to ensure reproducibility and comparability.

- **CIFAR:** A smaller open-source dataset suitable for rapid prototyping and stability testing found here: https://www.cs.toronto.edu/~kriz/cifar.html [15]. Two variants exist: CIFAR-10 (6000 images per each one of the 10 classes) and CIFAR-100 (100 classes with 600 images each).
- **ImageNet:** Open-source dataset from https://image-net.org/ [16]. Several variants of the dataset exist, ranging from the full dataset of around 1.3TB to a partial dataset of around 130Gb to Tiny ImageNet of only around 600Mb. This is a large dataset: 1. HARD TO OBTAIN - DO YOU HAVE THE DATA? 2. VERY LONG TO RUN. DO YOU HAVE ACCESS TO GPU?

Both sources of datasets have appeared in many visual transformer literature [10], [11], [13] and can be used as an exact comparison, not requiring a replication and retraining. For example, the previous work with Tokens-to-Token has evaluation metrics on CIFAR10, CIFAR100, ImageNet but don't necessarily have all metrics that our study is interested in. Depending on available resources and priority of resources, this study will use a combination of CIFAR-10, CIFAR-100, Tiny ImageNet, and any smaller form of ImageNet.

V. Expectations and Anticipated Challenges

A. Expected Results

We hypothesize that mHC-enhanced ViTs will:

- Demonstrate improved training stability, particularly in deeper architectures.
- Exhibit better-conditioned gradient norms across layers.
- Achieve comparable or modestly improved classification accuracy relative to base ViTs.
- Comparison between mHC to other extensions of ViTs like Tokens-to-Token is unclear, any comparable results (classification accuracy) is good given that stabilization is the primary focus.
- In the case of mHC-lite or KromHC, these extensions should maintain stability benefits with reduced computational overhead (measured in time to train, memory usage, etc).

B. Anticipated Challenges

Potential challenges include:

- Properly and efficiently implementing several models from prior research papers (mHC, Tokens-to-Token, KromHC, mHC-lite, etc.)
- Increased computational cost due to manifold projection operations.
- Ensuring fair and consistent hyperparameter tuning across model variants.
- Could be the case that stability and scalability benefits are not realized on small datasets like CIFAR and Tiny ImageNet (in which case a limit on available resources could limit the view on benefits).

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